# Multimedia Appendix 10. Study-specific risk of bias judgments.

| | Random sequence generation (selection bias) | Allocation concealment (selection bias) | Blinding of participants and personnel (performance bias) | Blinding of outcome assessment (detection bias) | Incomplete outcome data (attrition bias) | Selective outcome reporting | Other | Summary |
|---|---|---|---|---|---|---|---|---|
| Aittasalo 2006 [131] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Aittasalo 2012 [41] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Alsaleh 2016 [79] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Ashton 2017 [80] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Baker 2008 [81] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Barnes 2015 [42] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Barwais 2013 [82] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Bennett 2008 [155] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Butler 2004 [83] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Cadmus-Bertram 2019 [84] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Carr 2013 [148] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Coelho 2018 [43] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Compernolle 2015 [85] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Creel 2016 [86] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Croteau 2004 [129] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Croteau 2007 [130] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Cruz 2016 [44] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Dadaczynski 2017 [87] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| De Blok 2006 [88] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| De Greef 2010 [73] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| De Greef 2011 [89] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| De Greef 2011 [74] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Demeyer 2017 [90] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Dishman 2009 [91] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Dlugonski 2012 [45] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Duru 2010 [92] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Eakin 2014 [46] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Edney 2020 [47] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Engel 2006 [93] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Finkelstein 2016 [48] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Fischer 2019 [132] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Fjeldsoe 2010 [94] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Fjeldsoe 2015 [49] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Furber 2010 [50] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Gell 2015 [95] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Gill 2019 [96] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Glasgow 2012 [156] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Glynn 2014 [97] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Golsteijn 2018 [133] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Hardeman 2020 [98] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Harris 2018 [75] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Hornikx 2015 [99] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Hospes 2009 [100] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Houle 2011 [101] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Hultquist 2005 [102] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Izawa 2012 [103] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| James 2015 [51] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kangasniemi 2015 [40] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Katzmarzyk 2011 [104] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kawagoshi 2015 [105] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kendzor 2017 [134] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kernot 2019 [52] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Keyserling 2008 [135] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kim 2018 [136] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| King 2008 [137] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| King 2013 [106] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Koizumi 2009 [138] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Kolt 2012 [76] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Lane 2015 [149] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Li 2017 [139] | ● | ● | ● | ● | ● | ● | ● | High Risk |

| | Random sequence generation (selection bias) | Allocation concealment (selection bias) | Blinding of participants and personnel (performance bias) | Blinding of outcome assessment (detection bias) | Incomplete outcome data (attrition bias) | Selective outcome reporting | Other | Summary |
|---|---|---|---|---|---|---|---|---|
| Li 2020 [107] | + | + | - | + | + | + | + | Low Risk |
| Long 2013 [108] | ? | ? | - | + | - | ? | + | High Risk |
| Lynch 2019 [109] | + | + | - | + | + | - | - | Low Risk |
| Lyons 2017 [110] | + | + | - | + | + | - | + | High Risk |
| Maher 2015 [53] | + | + | - | + | + | + | + | Low Risk |
| Mailey 2010 [150] | ? | ? | - | + | + | ? | - | High Risk |
| Mansi 2015 [54] | ? | + | - | + | + | + | + | Low Risk |
| Martin 2015 [111] | ? | ? | - | + | + | + | - | High Risk |
| Maselli 2019 [55] | + | + | - | - | + | ? | - | High Risk |
| Maxwell-Smith 2019 [140] | + | + | - | + | + | + | + | Low Risk |
| Melville 2015 [112] | + | + | - | + | - | - | + | High Risk |
| Mendoza 2015 [113] | + | - | ? | + | + | + | + | Low Risk |
| Merom 2007 [114] | + | ? | - | - | + | ? | - | High Risk |
| Motl 2011 [151] | ? | + | - | - | + | ? | + | High Risk |
| Müller 2016 [56] | + | + | - | - | + | - | + | High Risk |
| Murawski 2019 [57] | + | + | - | - | + | ? | + | High Risk |
| Mutrie 2012 [58] | + | + | - | + | + | + | + | Low Risk |
| Nolan 2017 [59] | + | + | - | + | - | + | - | High Risk |
| Oliveira 2019 [60] | + | + | - | + | - | + | + | High Risk |
| Paul 2016 [115] | - | - | - | + | + | ? | - | High Risk |
| Pekmezi 2017 [141] | ? | ? | - | + | + | - | - | High Risk |
| Pinto 2013 [61] | ? | ? | ? | - | + | - | + | High Risk |
| Pinto 2015 [62] | ? | + | - | + | + | ? | + | Low Risk |
| Poirier 2016 [116] | + | - | - | + | - | + | ? | High Risk |
| Pope 2018 [117] | + | - | - | + | - | ? | - | High Risk |
| Prestwich 2009 [142] | + | ? | - | - | - | ? | - | High Risk |
| Prestwich 2010 [118] | + | + | - | - | + | ? | - | High Risk |
| Reijonsaari 2012 [119] | + | + | + | - | - | + | + | High Risk |
| Ribeiro 2014 [63] | + | + | - | + | - | + | - | High Risk |
| Roos 2014 [120] | + | + | - | + | + | ? | - | High Risk |
| Rowley 2019 [121] | ? | - | - | + | - | ? | - | High Risk |
| Samuels 2011 [143] | ? | ? | - | + | - | ? | - | High Risk |
| Schwerdtfeger 2012 [154] | ? | ? | - | + | + | ? | - | High Risk |
| Sharp 2016 [144] | + | ? | - | - | - | ? | - | High Risk |
| Simons 2018 [64] | + | + | - | + | - | + | + | High Risk |
| Spence 2009 [122] | ? | ? | - | - | + | ? | - | High Risk |
| Stacey 2016 [65] | + | - | - | + | - | - | - | High Risk |
| Sugden 2008 [152] | + | ? | - | + | + | ? | - | High Risk |
| Suggs 2013 [66] | + | ? | + | - | - | ? | - | High Risk |
| Tabak 2014 [123] | + | + | - | + | + | + | + | Low Risk |
| Talbot 2003 [67] | + | ? | - | + | + | ? | - | High Risk |
| Ter Hoeve 2018 [68] | + | + | - | + | + | - | - | High Risk |
| Thorndike 2014 [124] | + | + | ? | + | + | ? | ? | Low Risk |
| Thorsteinsen 2014 [153] | ? | + | - | - | + | ? | + | High Risk |
| Tudor-Locke 2004 [69] | ? | ? | - | + | - | ? | - | High Risk |
| Unick 2012 [145] | ? | ? | - | + | + | ? | - | High Risk |
| Vallance 2008 [70] | + | + | - | - | - | + | + | High Risk |
| Vallance 2016 [125] | + | + | - | + | + | ? | - | High Risk |
| Van Blarigan 2019 [126] | + | - | - | + | + | + | + | High Risk |
| Vandelanotte 2018 [146] | + | + | + | - | - | + | + | High Risk |
| Van der Weegen 2015 [71] | + | + | - | + | + | + | - | High Risk |
| Van Hoye 2018 [72] | + | ? | - | + | + | + | - | High Risk |
| Warren 2014 [127] | + | + | - | + | - | + | - | High Risk |
| Wijsman 2013 [147] | + | + | - | + | + | + | + | Low Risk |

| | Random sequence generation (selection bias) | Allocation concealment (selection bias) | Blinding of participants and personnel (performance bias) | Blinding of outcome assessment (detection bias) | Incomplete outcome data (attrition bias) | Selective outcome reporting | Other | Summary |
|---|---|---|---|---|---|---|---|---|
| Wyke 2019 [77] | ● | ● | ● | ● | ● | ● | ● | High Risk |
| Yamada 2012 [128] | ● | ● | ● | ● | ● | ● | ● | Low Risk |
| Yates 2017 [78] | ● | ● | ● | ● | ● | ● | ● | Low Risk |

*Note:* Table depicts risk of bias assessment per bias domain and an overall summary assessment. Summary assessment ranked as high if one domain other than "Blinding of participants and personnel" is ranked high.

●= low risk, ●= unclear risk, ●= high risk.